



Design of a Silicon Nitride Photonic CNOT Gate For Linear Optical Quantum Computing

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Project Summary



This project presents the design of a proof-of-concept photonic CNOT gate for integrated quantum computing.

The aim is to realise a key two-qubit quantum logic operation using an on-chip photonic platform, supporting a route toward compact, low-cost, and scalable quantum technologies.

The design uses silicon nitride waveguides at 780 nm, with supporting components such as grating couplers, MMIs and MZIs to create a practical integrated-photonics implementation of a quantum gate.

The scope of this project is limited to on-chip photonic components. Off-chip propagation, including light before coupling onto the chip and after coupling from it, is not considered.

Design Rationale



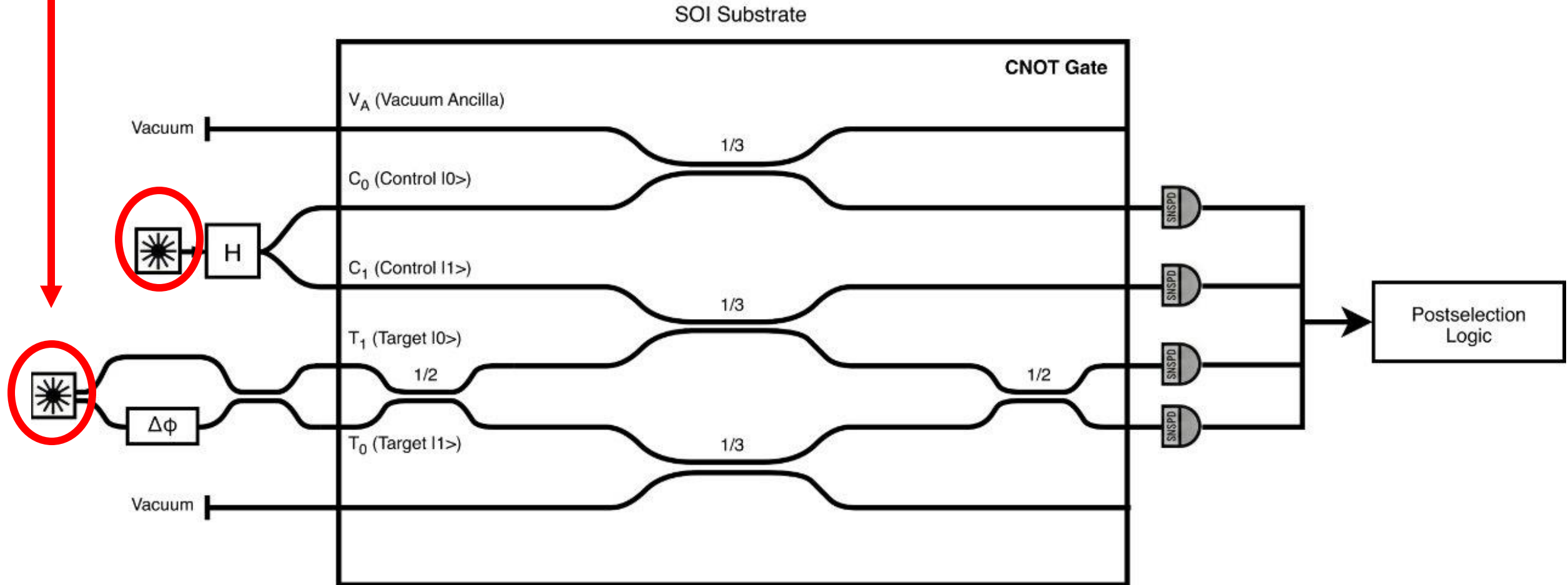
- The device was designed for operation at a **wavelength of 0.78 μm** (780 nm). This choice was motivated by the large volume of existing work on 1.5 μm photonic couplers, while the 780 nm region remains comparatively less explored. The project therefore aims to contribute to a less saturated area of photonics research.
- The **waveguide material** was chosen as **Silicon Nitride (Si_3N_4)** due to its greater suitability for 0.78 μm than standard silicon photonics
- **Grating Couplers** were included as the on-chip optical interface, allowing light to be coupled vertically between optical fibre and the photonic chip.
- **MMIs (Multimode Interferometers)** rather than DCs (Directional Couplers) due to their in general more fabrication tolerance and provide a more robust 50:50 split, making them better suited to a proof-of-concept integrated photonic device
- **MZIs (Mach- Zender Interferometers)** were chosen because they provide a practical way to control interference and optical phase on-chip, allowing the required quantum state transformations for the photonic CNOT gate

The project was split into three sections

- Grating Couplers – Morgan McCabe
- 50:50 MMI – Casey Beck
- 33:66 MMI – Oliver French



The Grating Couplers



Design Parameters to optimize

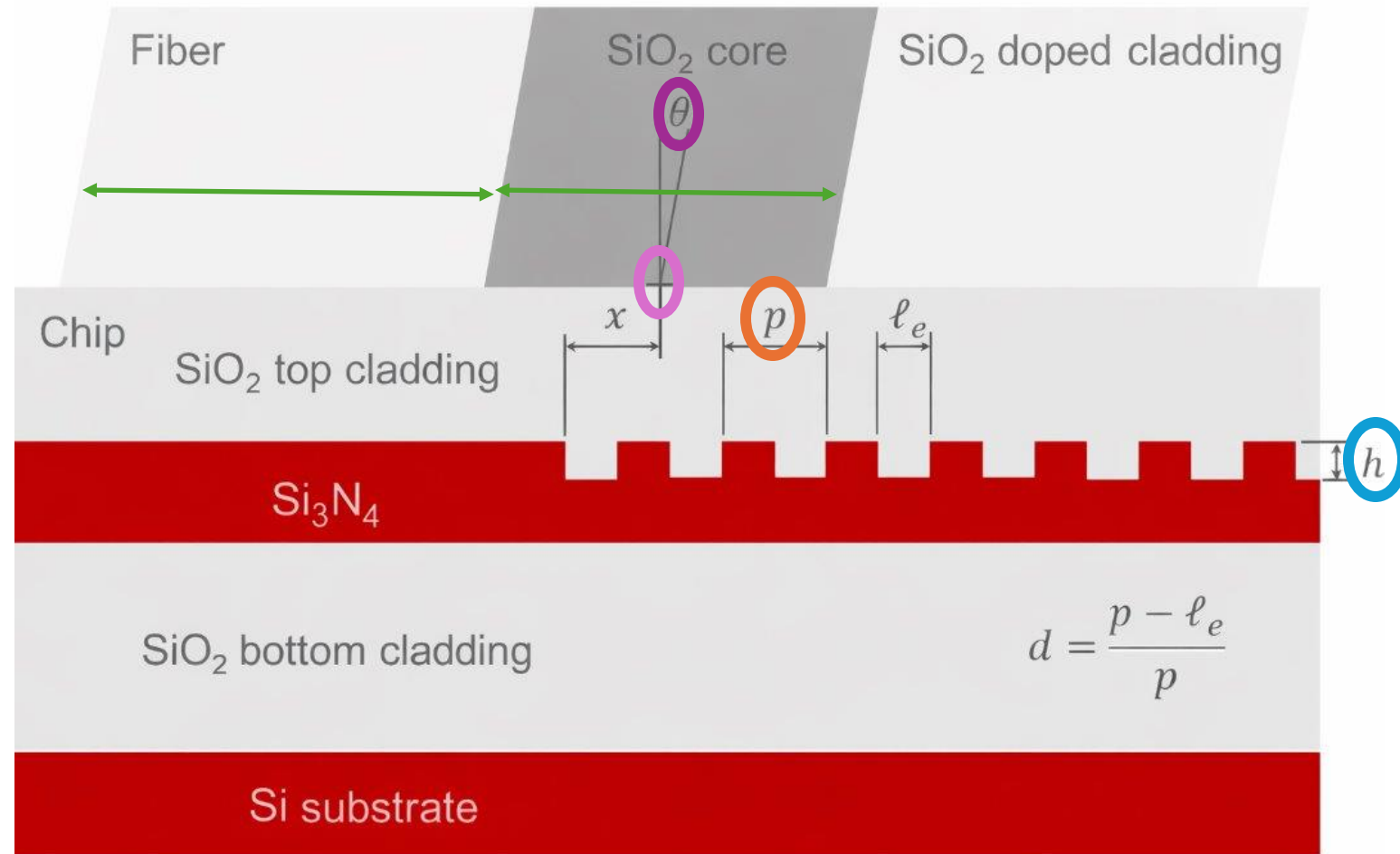


High priority

- **Fibre angle** – the angle of the fibre to the normal (the grating region) = θ
- **Etch depth** – the height of the teeth / the depth of the grating = h
- **Pitch** - Pitch is the size of one period, which is the tooth width + gap width = P
- **Duty cycle** - the fraction of how much of one pitch is occupied by the tooth. It lies between 0 – 1 = $P - L_e$
- **Fibre core Position** – the physical area it emits / lies on the grating area

Low Priority

- **Fibre core/cladding diameter** – the diameter of the fibre that is vertically directing the light onto the grating area
- **Period number** = grating length / pitch = number of x
- **Grating width / length** – I.e the overall grating shape / length. Could be rectangular, or more cone shaped



Initial Parameter Setup via Sources and estimations



Sourced :

- **Fibre Angle** = I chose 10°
 - (Kumar Selvaraja et al., n.d.) “Both the fibres were positioned at an angle of 10 degrees.”
 - (Chen, J. Thomson and Crudginton, 2017) “Fibres were aligned to the surface of the SOI wafer with a tilting angle of $\theta = 10$ off normal”
- **Core/Cladding Diameter** = 9um and 50um
 - (OPTICAL FIBERS Features, n.d.)
- **Core/Cladding Refractive Index** = 1.44427 and 1.43482
 - (Optica.org, 2026)

Estimations :

- grating phase matching equation (**Pitch**) =
$$\frac{m\lambda}{(n_{eff} - n_{clad}) * (\sin \theta)}$$
 - m = diffraction order (1)
 - λ = wavelength we are coupling (0.78um)
 - **Neff**= effective index of optical mode (1.6496)
 - **Nclad** = effective index of cladding (1.44)
 - **Θ** = fibre angle to normal (grating) (10)
 - We get the value 0.558um for initial pitch
- **Etch Depth** = max height can be 0.3um , so a safe initial value can be 0.15um.
- **Duty Cycle** = This value is the fraction of how much of one pitch is occupied by the tooth. It lies between 0 – 1. Initially we will set it to 0.5.
- **Grating Width/Length and Fibre position** = Set to 25um in x plane, this isn't a very important factor, as long as it's big enough to fit the entire fibre core. Therefore the fibre position can be set to around 12.5um in the x plane

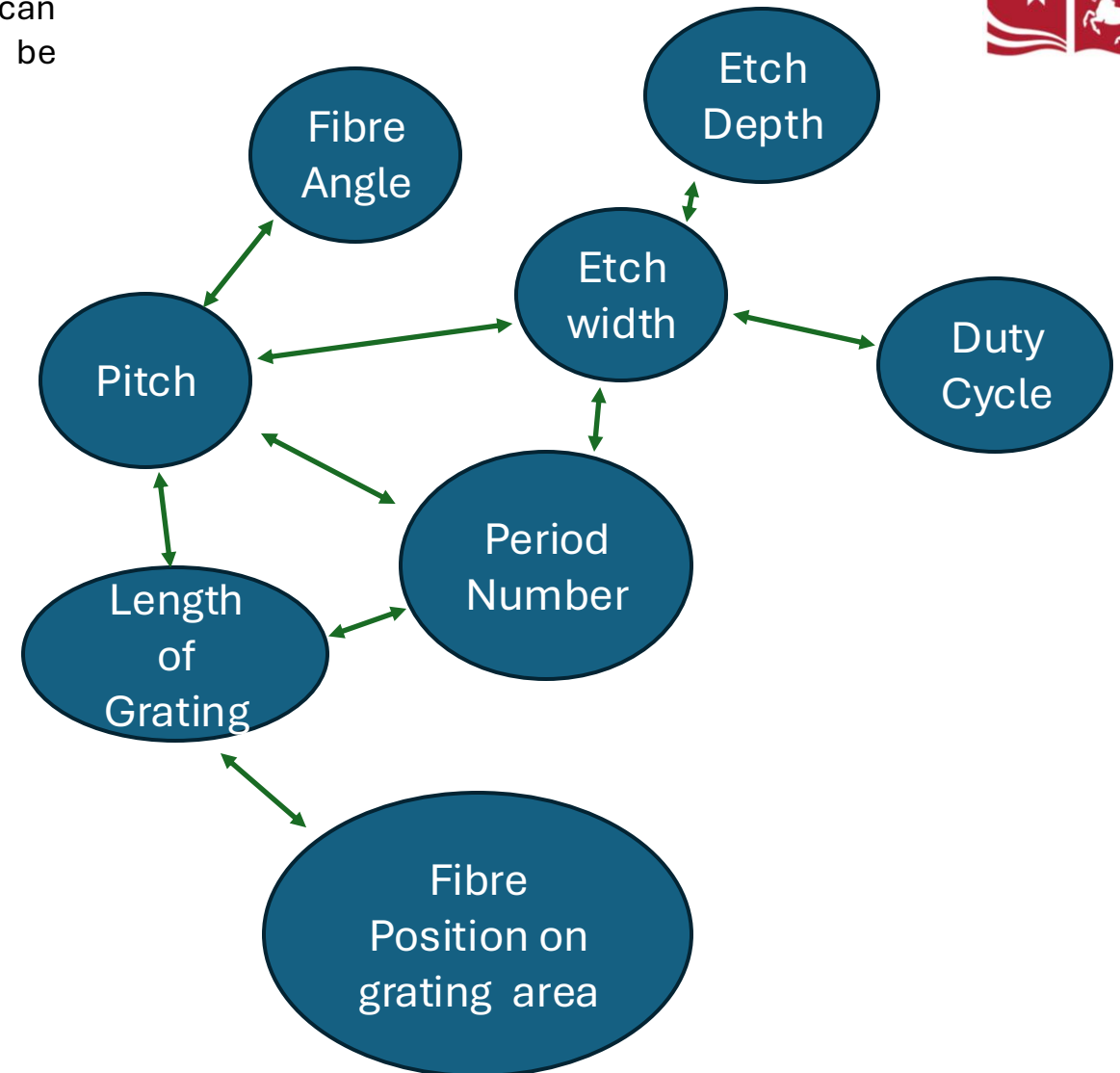
Parameter Relationship



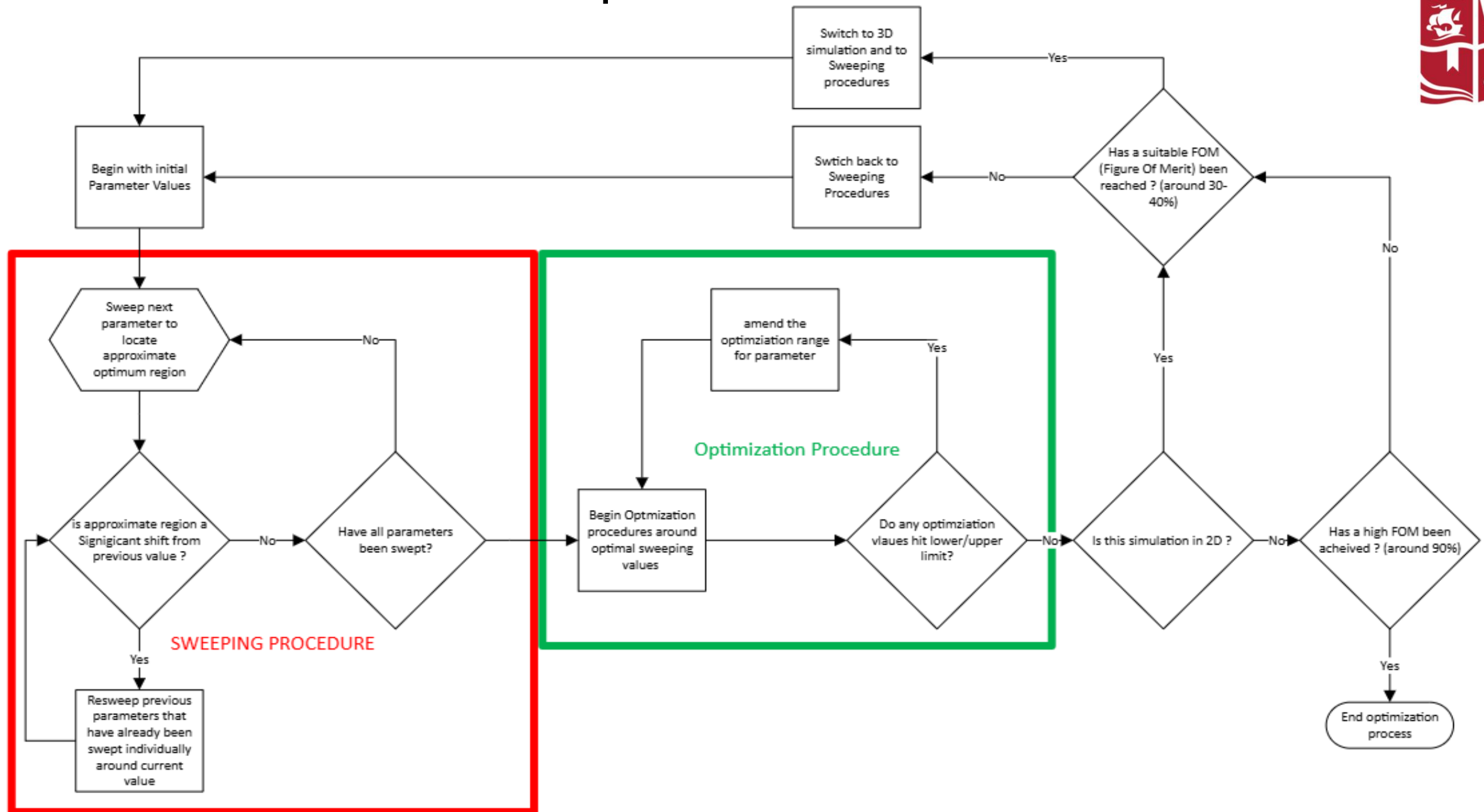
Due to the optimizable parameters interdependence, changing one parameter can shift the optimal values of the others. As a result, the design process must be iterative rather than strictly linear.

Main Linked Parameter Groups

- Phase-matching Group
 - Pitch
 - Fibre Angle
 - Wavelength (Fixed)
 - Effective index of core (Fixed)
 - Effective index of cladding (Fixed)
- Scattering Group
 - Duty Cycle
 - Etch Depth
 - Etch/tooth Width
 - Waveguide thickness (Fixed)
- Length Group – Becomes Important in 3D simulation
 - Number of periods
 - Grating area length (x plane)
 - Fibre x-position



Optimization Process

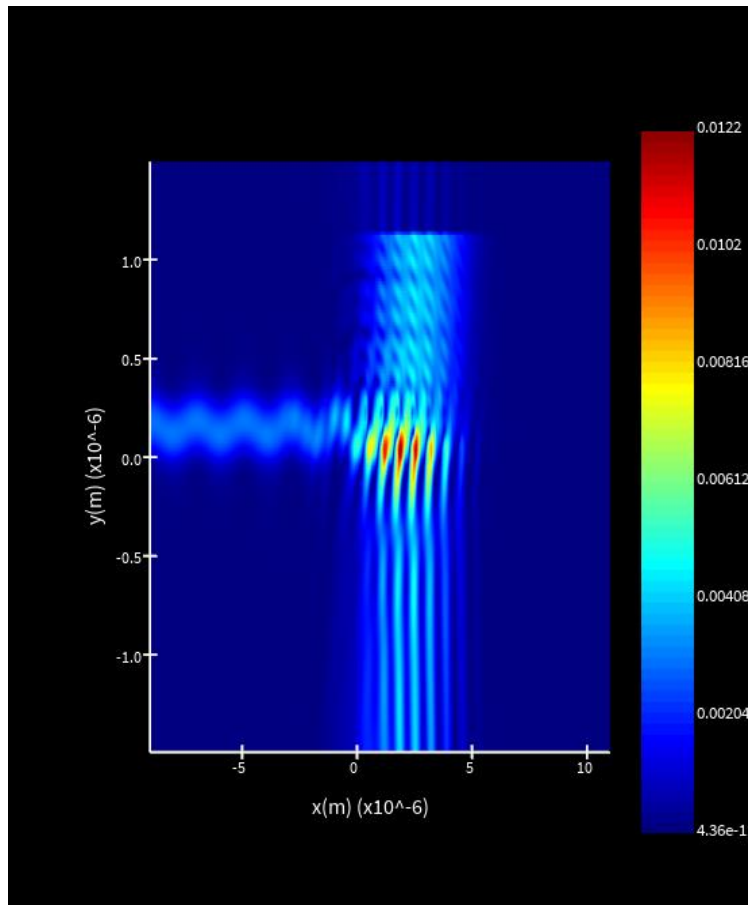


Processing 2D Data

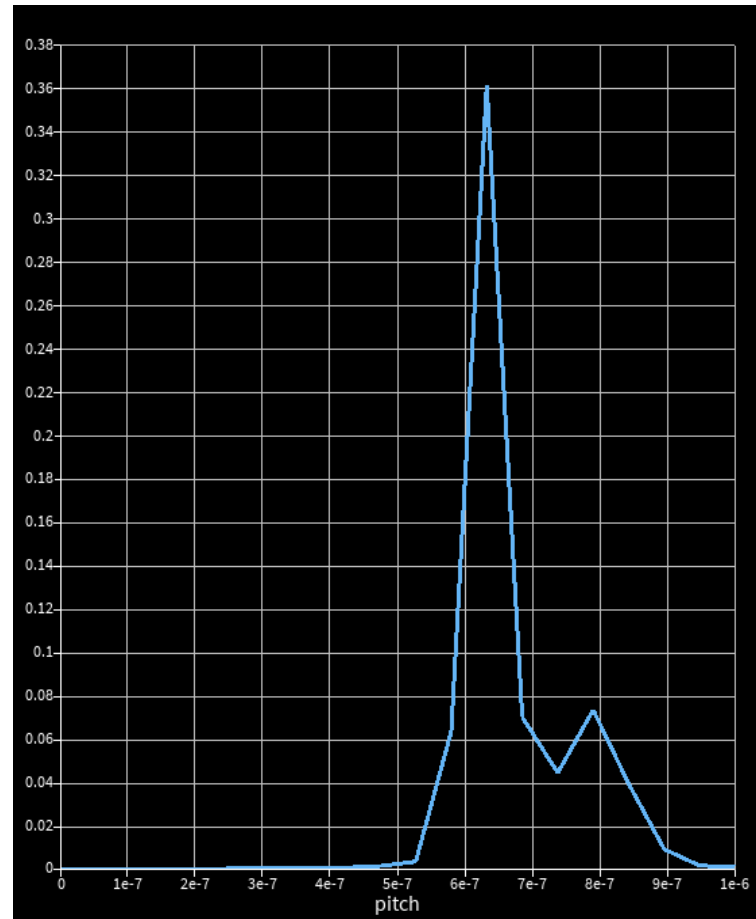


Sweeping procedures is more than often singular for each parameter, whereas optimizing is usually multiple parameters at a time

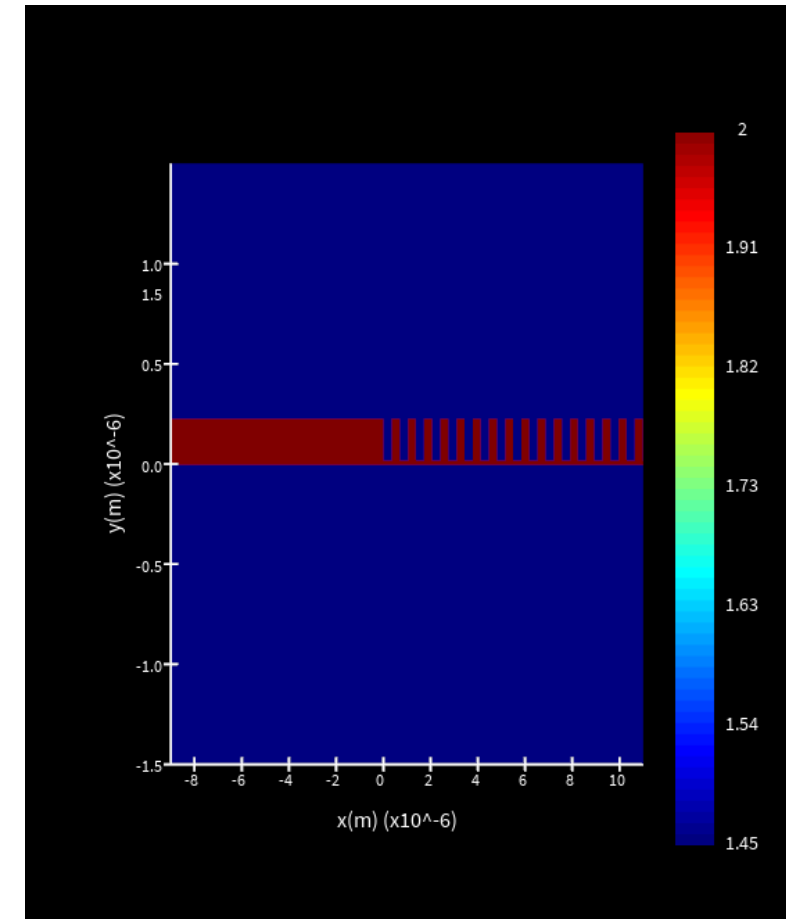
Frequency monitor before any optimizing, monitors the designated wavelength (0.78um), visualizing the coupled light



Here is an example of a sweep procedure for the pitch value. It peaks at the value 0.631579. Increasing the FOM to 0.36 (36%).



Refractive index monitor, measuring the index of the grating material and the top/bottom substrate .



2D Simulation to 3D Simulation

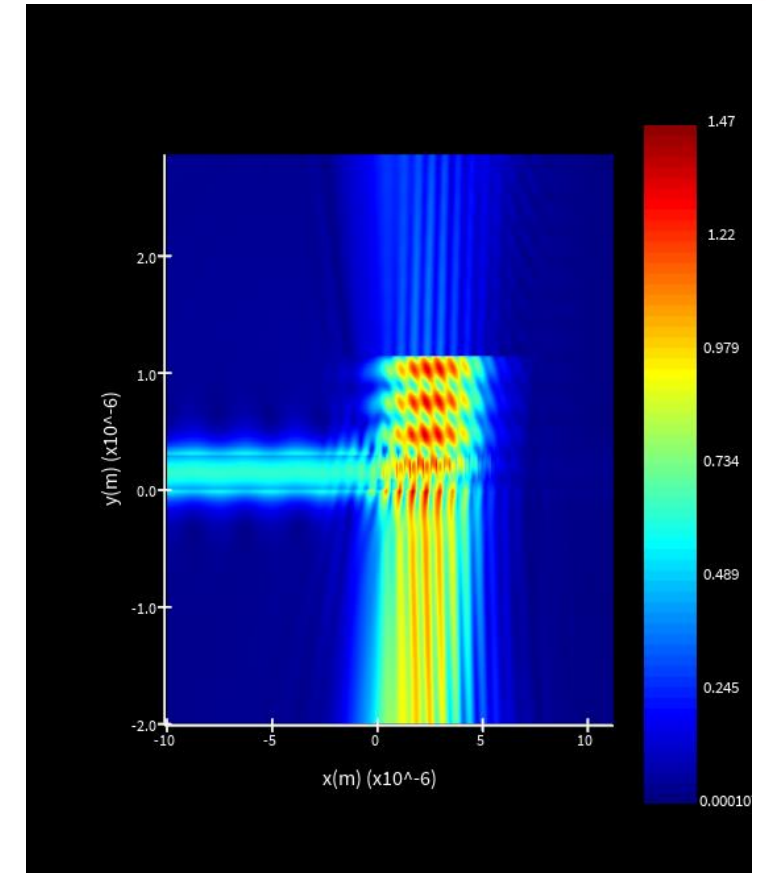
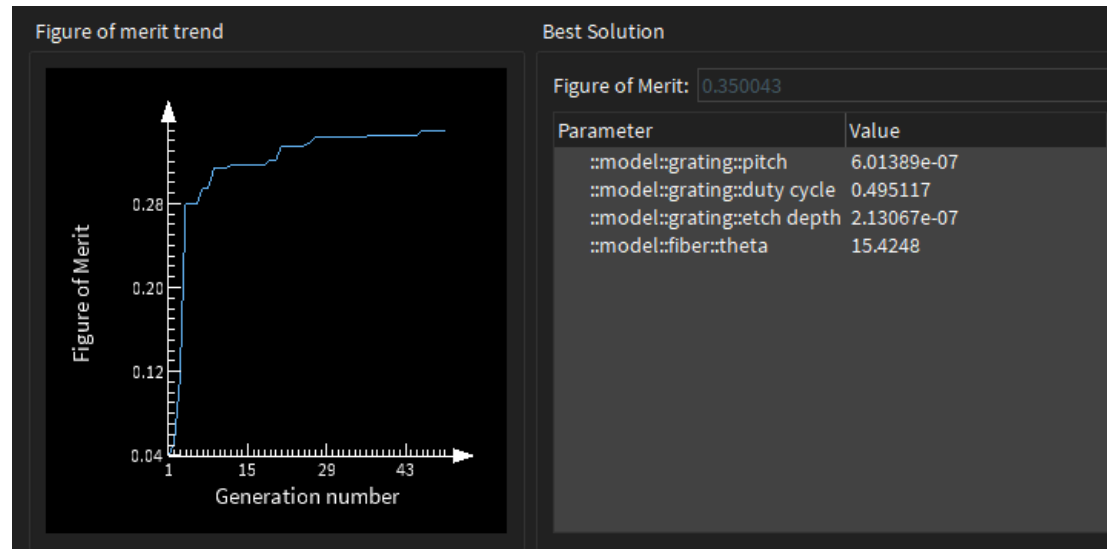


Iteration Number	Pitch	Etch Depth	Duty Cycle	Fibre Angle	FOM
1	0.558	0.15	0.5	10	0.0728213
...					
22	0.6	0.2	0.484114	18	0.2
23	0.631579	0.2	0.484114	18	0.31
24	0.631579	0.205203	0.484114	18	0.32
25	0.631579	0.205203	0.473684	18	0.30
26	0.631579	0.205203	0.484114	17.8947	0.29

Perfecting around values for iteration 24

FOM = 0.35

Suitable to move to 3D



Frequency monitor graph at iteration 24



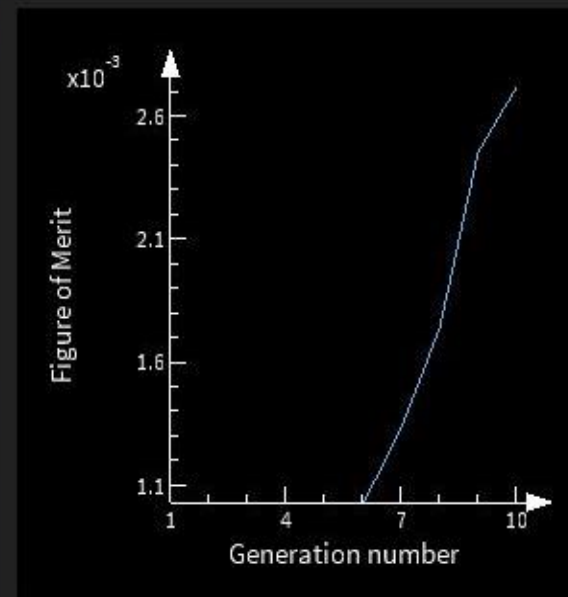
3D Simulation

Iteration Number	Fibre x position	Duty cycle	Pitch	Etch Depth	Fibre Angle	FOM
1	N/A	0.495117	0.601389	0.213067	15.4248	0.0000109749
36	4.03024e-06	0.181268	1.04399e-06	0.15e-06	8.20049	0.00274478
37	4.54622e-06	0.2	1.04353e-06	0.147718	7.41282	0.002..
38	4.04045e-06	0.15	1.05e-06	2.19329e-07	8.5	0.00254392
39	3.1592e-06	0.4031111	5e-07	9.23682e-08	5	0.016623
40	4.18654e-06	0.365791	5.11408e-07	1.19062e-07	8.95381	0.0181321

The blue are where the sweeps / optimizations procedures hit the lower limit when searching for optimal value, and red is hitting the upper limit.

Ran into incredibly low FOM values when moving into 3D, no matter how many iterations of Sweeps and optimizations were ran

Figure of merit trend



Best Solution

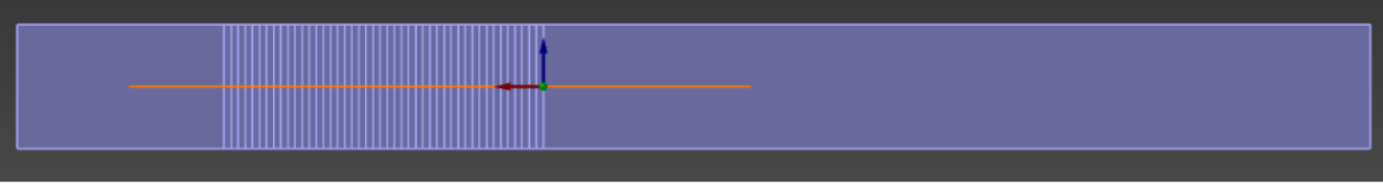
Figure of Merit: 0.00271684

Parameter	Value
::model::fiber::grating target length	4.03024e-06
::model::grating::duty cycle	0.2
::model::grating::pitch	1.04353e-06
::model::grating::etch depth	1.47718e-07
::model::fiber::theta	7.41282

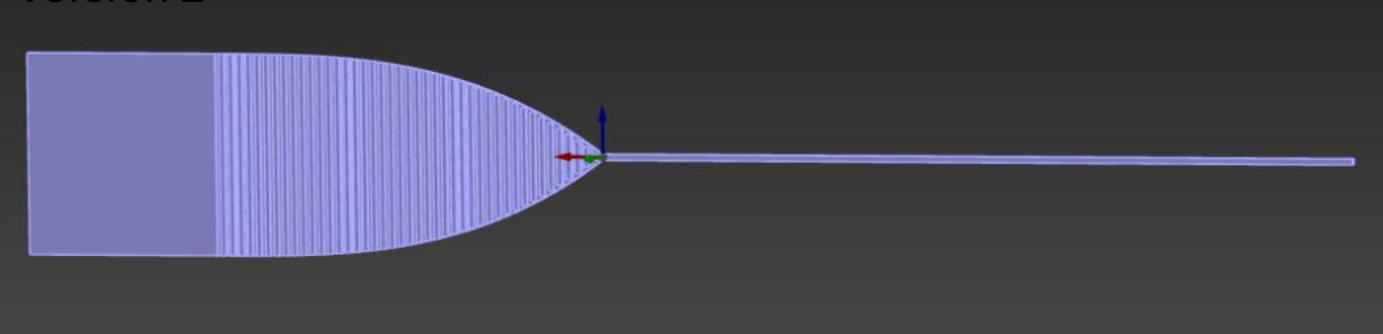
Grating Shape



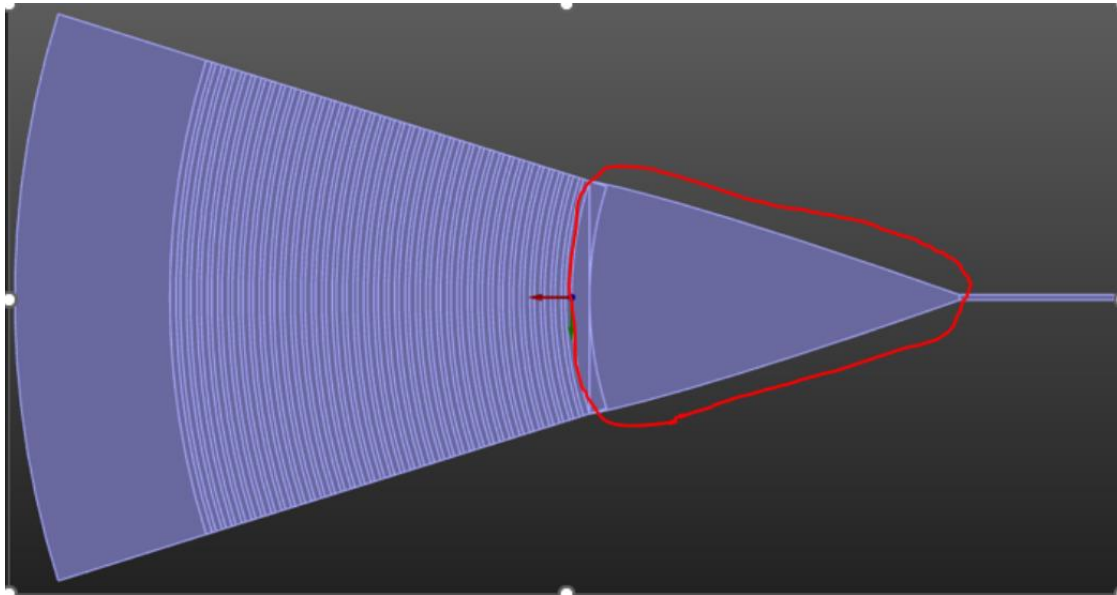
Version 1



Version 2



Version 3 (final version)



In 3D, the FOM stayed very low (**~0.002–0.015**, see graph above) even after repeated sweeps and optimization.

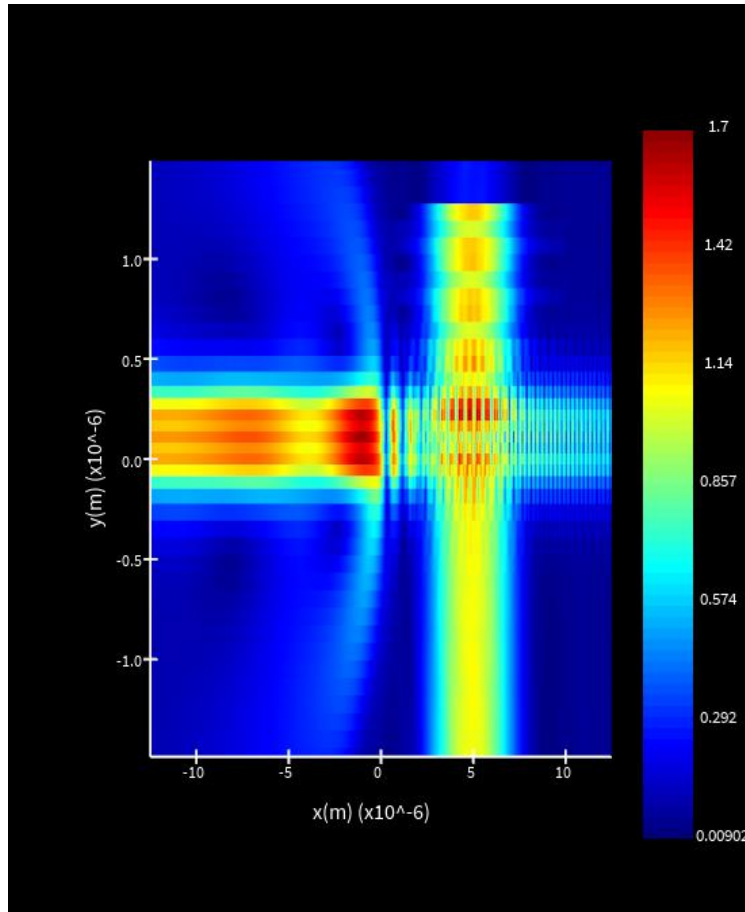
This suggested the issue was not just parameter choice, but the overall geometry.

Literature showed that many efficient designs use a **cone-shaped grating region** much wider than the waveguide.

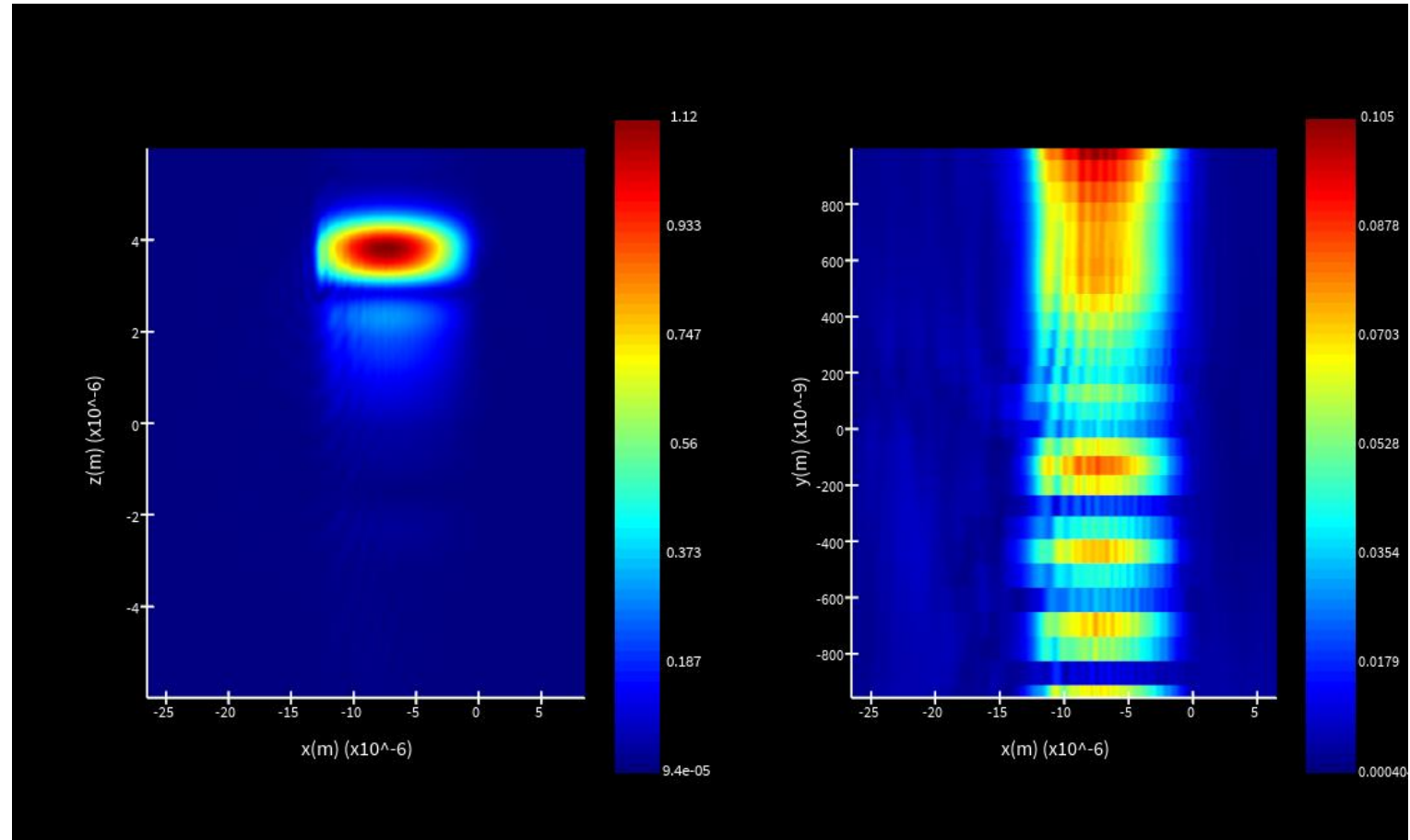
- (Almeida et al., 2024)
- (Ansys Optics, 2021)

To address this, I introduced a **broader grating region** and a **taper (red circle)** between the waveguide and grating, to create a smoother mode transition.

3D Simulation



Frequency Monitor graph for Version 2 grating shape (previous slide)



Frequency Monitor graph for Version 3 grating shape (previous slide). The left shows the light coupled in the fibre travelling down the y axis. The right shows the light coupling in the waveguide in the xy plane

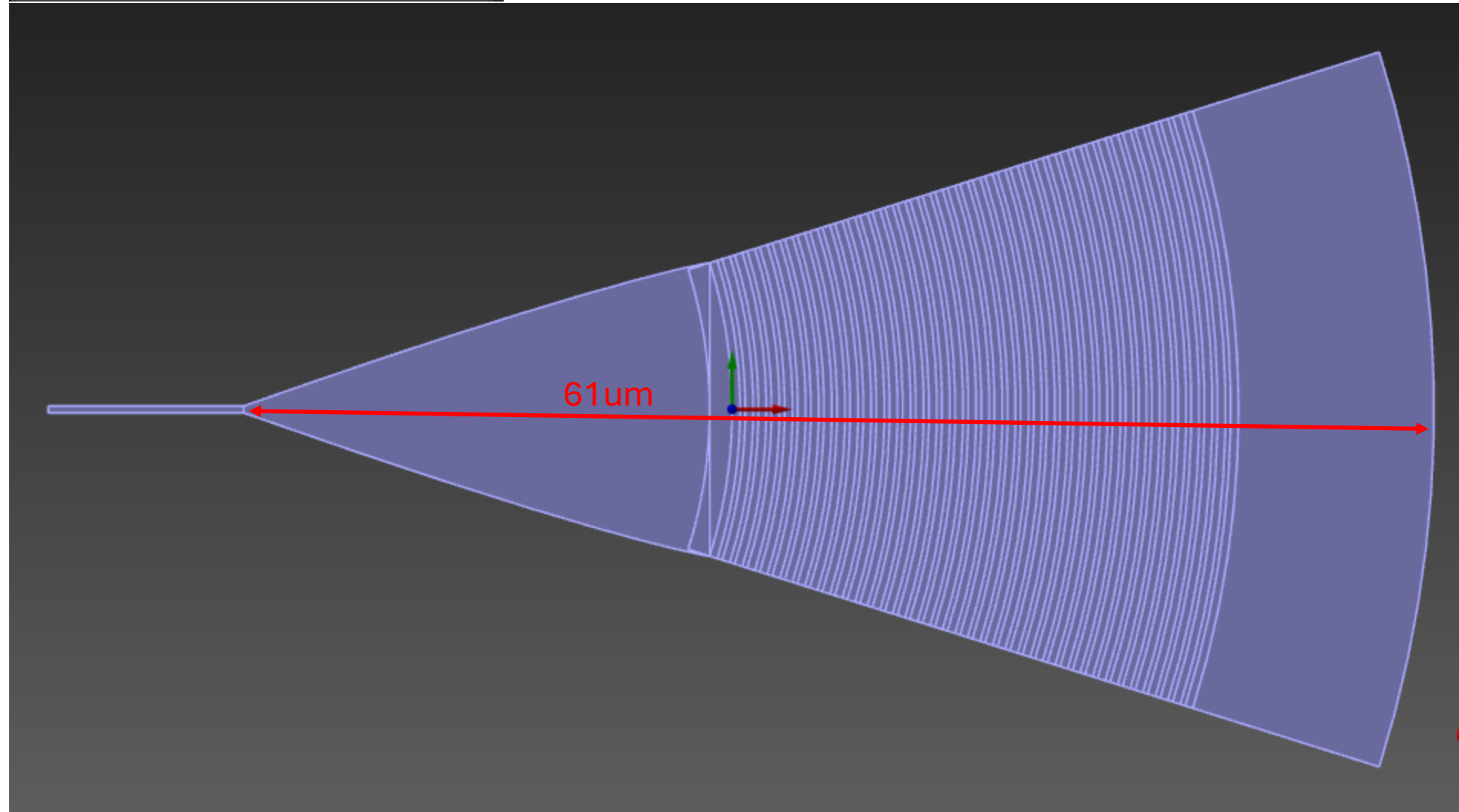


Final Grating Component

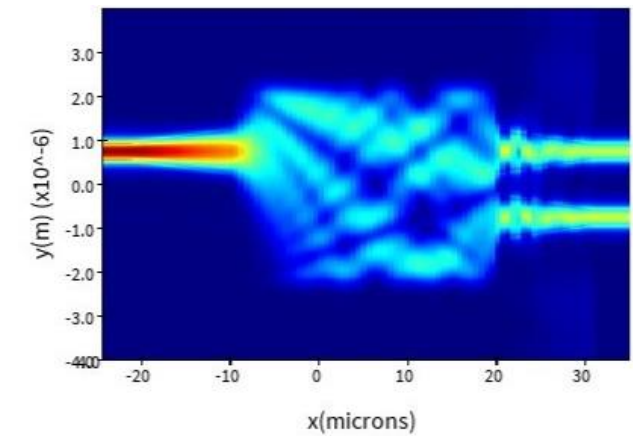
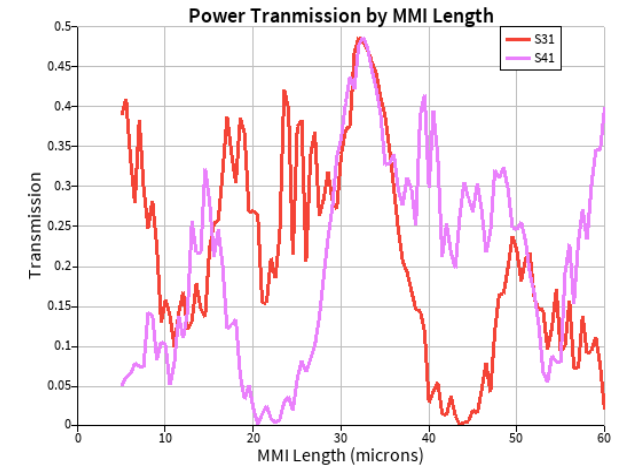
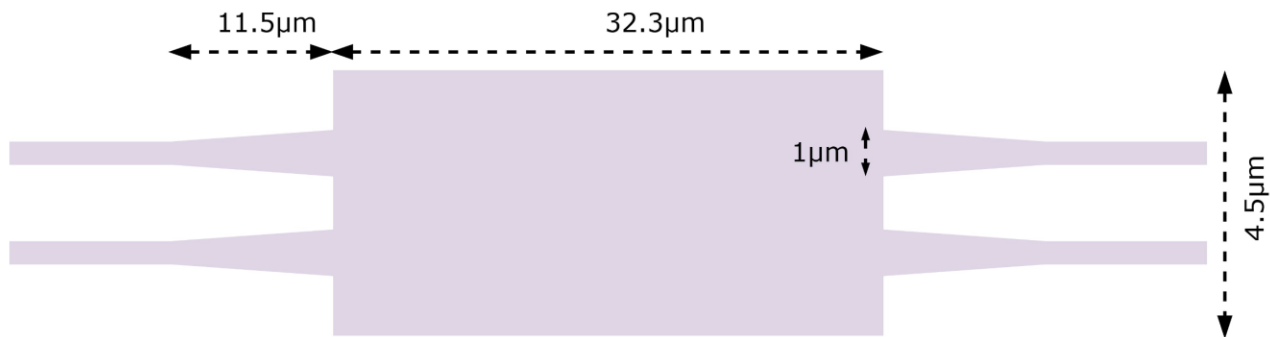
With a 98.5% FOM , with the set parameters (in um):

- Pitch (0.6713)
- Etch Depth (0.1)
- Fibre Angle (10 degrees)
- Grating length (25)
- Taper Length (10)
- Duty Cycle (0.3992)
- Period Number (38)
- Curve of the gratings/ radius (25)
- Width of the taper length (15)
- Length of extra waveguide (10)

Name	Dimensions/Value
 T	0.985274



50:50 Multi-mode Interferometer (MMI) Design



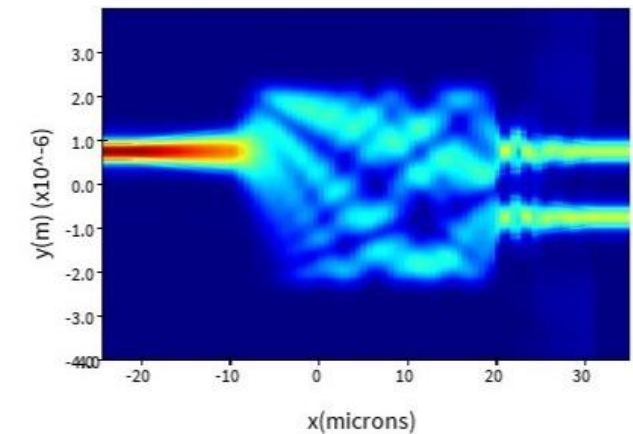
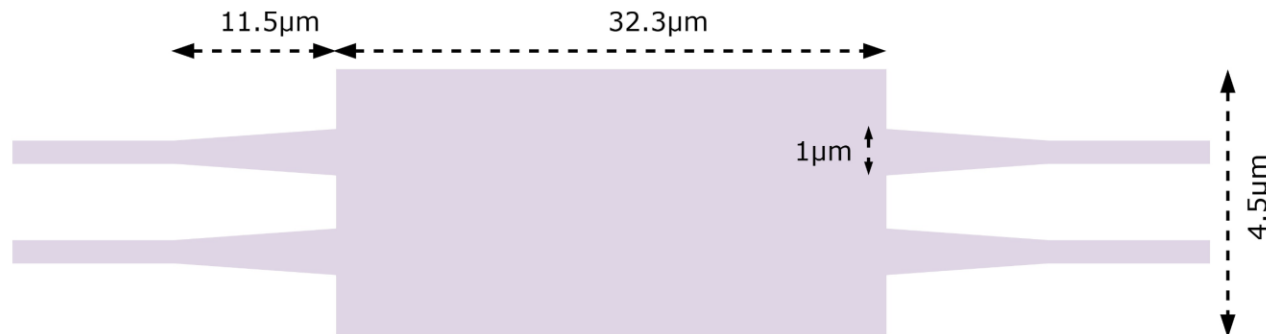
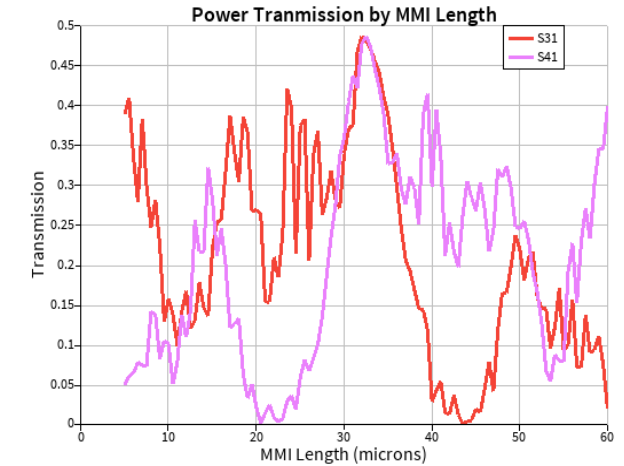
[x]

50:50 Multi-mode Interferometer (MMI) Design



Following a thorough collation and review of standard 1/2 MMI design practices, **a set of adjustable parameters were defined:**

- MMI Core Width (y)
- MMI Core Length (x)
- Taper Mouth Width (y)
- Taper Length (x)
- Taper Geometry (qualitative)



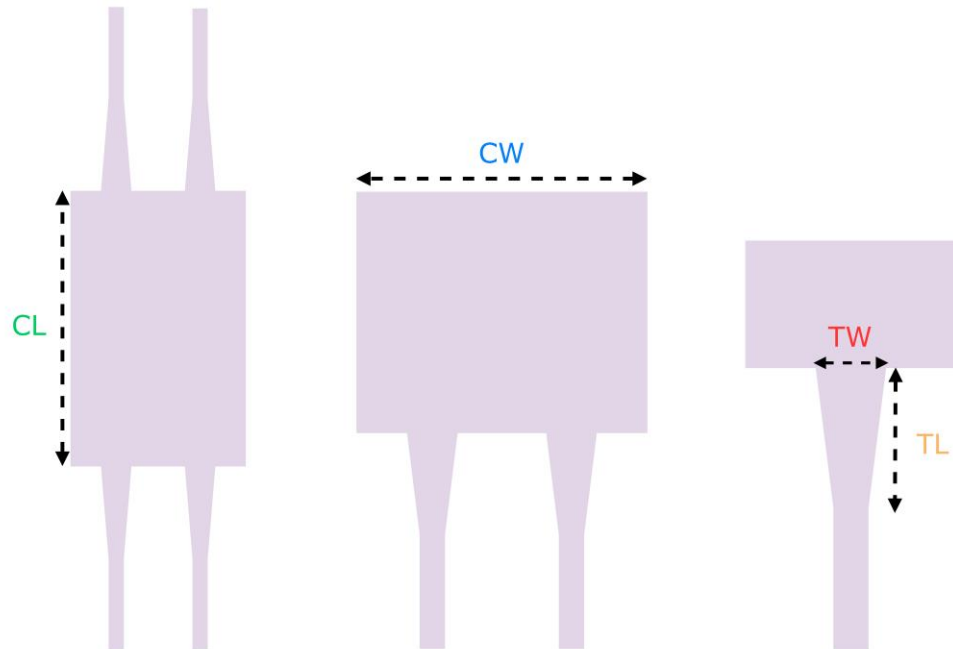
[x]

50:50 Multi-mode Interferometer (MMI) Design

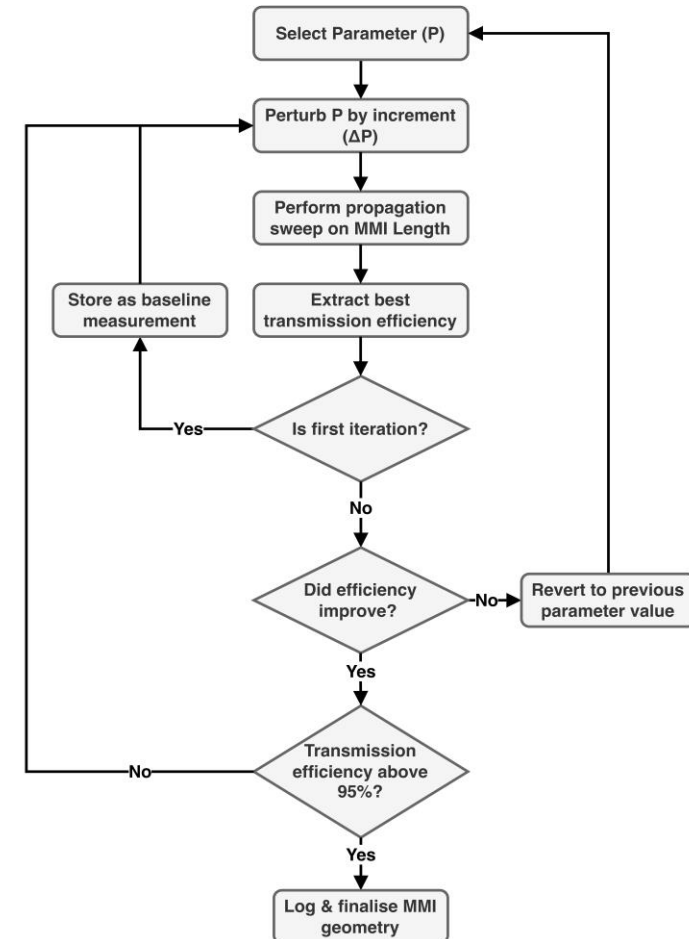


The defined parameters were then iteratively adjusted to achieve the final efficiency figure:

Parameter	Range (μm)	Selected Value (μm)
MMI Core Length (CL)	5-100	32.3
MMI Core Width (CW)	3-7	5.5
Taper Mouth Width (TW)	3	5
Taper Length (TL)	0.6-1	1



Process flow devised for MMI design



50:50 Multi-mode Interferometer (MMI) Design

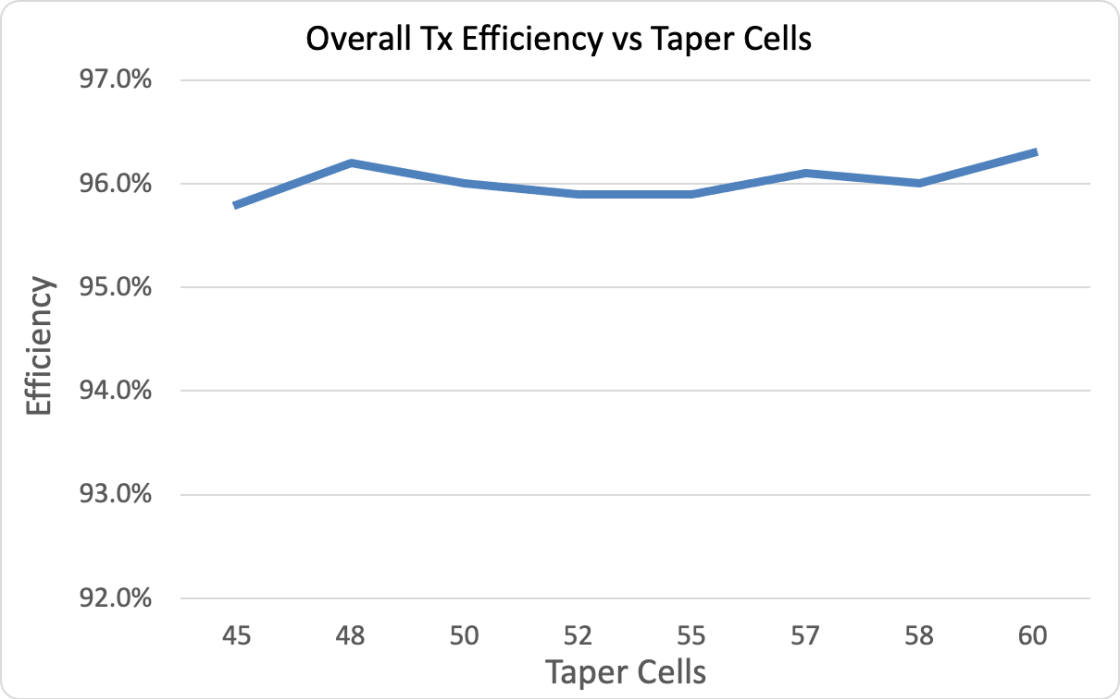


After achieving the desired efficiency figure, convergence testing was conducted to ensure that the value was not subject to inaccuracies due to simulation settings.

Base runs were conducted with the following EME parameters:

Group Span	Cells	Modes
1 – Straight Waveguide In	1	10
2 – Taper In	45	60
3 – MMI Core	1	150
4 – Taper Out	45	60
5 – Straight Waveguide Out	1	10

Cells were increased in increments of **2** from **45** to **60**. No significant performance improvements were noted, which indicates that convergence had already been achieved on base runs.



50:50 Multi-mode Interferometer (MMI) Design

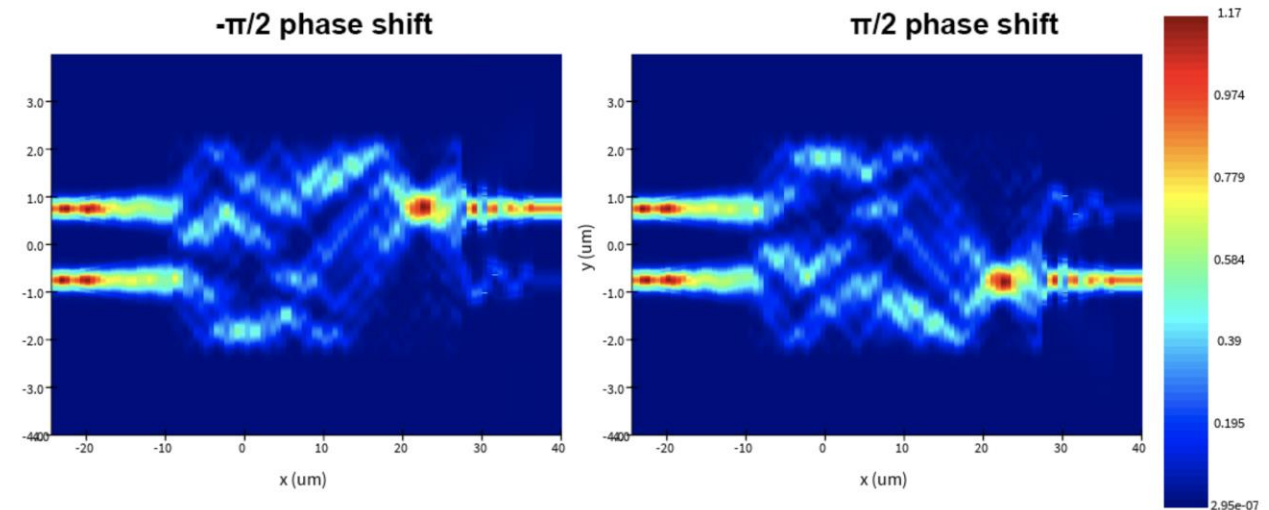


Finally, the MMI performance was validated by simulating **two-port operation**. This characteristic is essential for nominal operation when constructing a Mach-Zehnder Interferometer (MZI) from MMIs.

A small amount of power coupling was observed due to the proximity of the taper sections, though

[x]

50:50 MMI Two-Port Operation

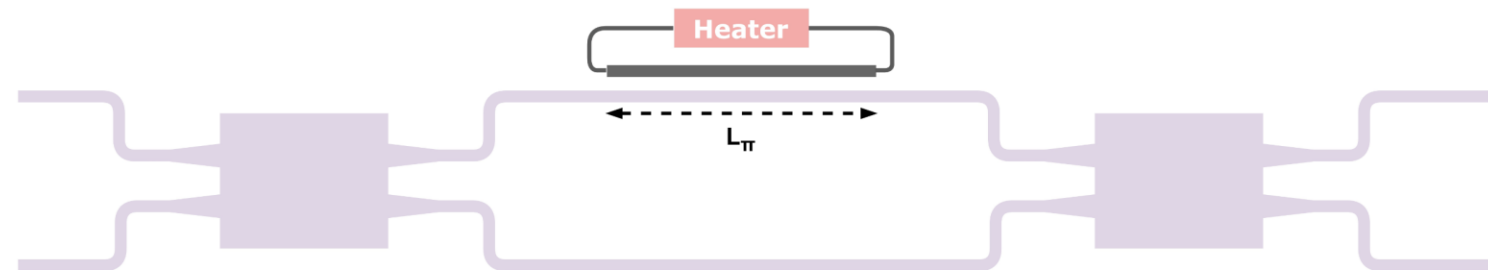


Mach-Zehnder Interferometer (MZI) Design



The MZI design involved a phase shifting section sandwiched between two 1/2 MMI couplers.

CORNERSTONE design rules necessitated a TiN thermo-optic heater for phase modulation.



Mach-Zehnder Interferometer (MZI) Design

Simulations were conducted on Lumerical HEAT to calculate the required heater dimensions and spacing between waveguide arms:

